Bachelorproefvoorstelling van grafenkleuringen

A computational approach for variants of coloring planar graphs

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Overzicht

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Onderzoeksprobleem

- Hoe kunnen we een efficiënt algoritme voor het kleuren van vlakke grafen maken, zodat we hiermee vermoedens kunnen testen?
- Knoopkleuringen van vlakke grafen
- Vermoedens rond chromatische getallen weerleggen of ondersteunen door computationeel bewijs

Grafenkleuringen

- Proper colouring

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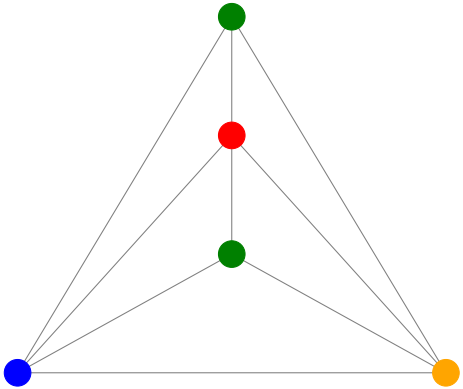

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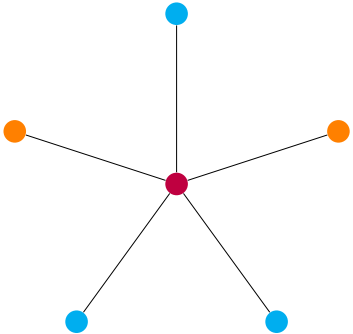

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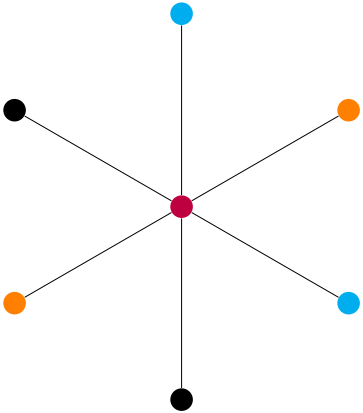

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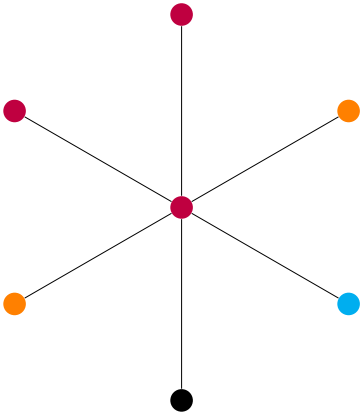

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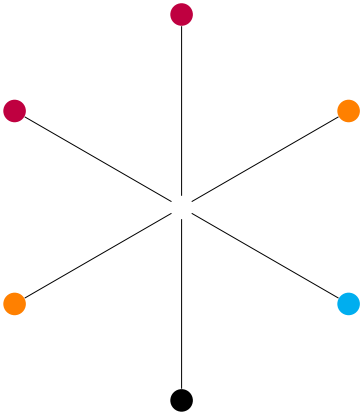

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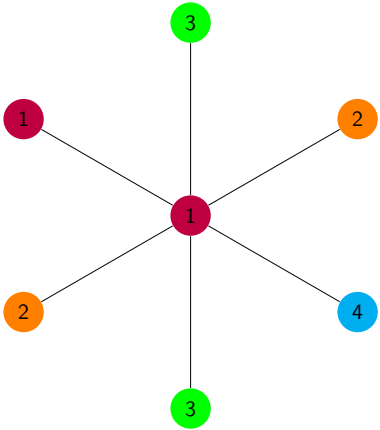

Toepassingen

- Draadloze netwerken: versturen signaal van zendmast zonder interferentie
  $\rightarrow$ Conflict free colouring
- Vertex ranking: kleuring waarbij paden tussen knopen met zelfde kleur een knoop bevatten met grotere kleur
  $\rightarrow$ Unique-Maximum colouring
  - Toepassingen bij parallel factoriseren matrices en meer

Naamgeving

- Proper $\rightarrow$ p
- Improper $\rightarrow$ i
- Open neighbourhood $\rightarrow$ o
- Closed neighbourhood $\rightarrow$ c
- pCFo, iUMo, iUMc, pUMo, iCFc, ...
- $2 \times 2 \times 2 + 2 = 10$ kleuringen

Chromatische getallen

- $\chi(\mathcal{P})$, $\chi_O(\mathcal{P})$, $\chi_{pCFo}(\mathcal{P})$, ... ($\mathcal{P}$ de klasse van vlakke grafen)
- Per kleuring
- Maximaal aantal kleuren, binnen klasse van vlakke grafen, bij minimale kleuringen van die grafen

Literatuur

- Brengen resultaten samen en bewijzen nieuwe grenzen[1,2,3]
- Formuleren vermoedens[1,2]

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[1] Igor Fabrici e.a. *Proper conflict-free and unique-maximum colorings of planar graphs with respect to neighborhoods*. Feb 2022. DOI: 10.48550/arXiv.2202.02570

[2] Mirko Petruševski en Riste Škrekovski. "Colorings with neighborhood parity condition". In: *Discrete Applied Mathematics* 321 (2022), p. 385–391. ISSN: 0166-218X. DOI: https://doi.org/10.1016/j.dam.2022.07.018. URL: https://www.sciencedirect.com/science/article/pii/S0166218X22002803

[3] Jan Petr en Julien Portier. *Odd chromatic number of planar graphs is at most 8*. Jan 2022. DOI: 10.48550/arXiv.2201.12381

Vermoedens

Odd colouring

Bewezen grenzen:

$5 \leq \chi_{\mathrm{O}}(\mathcal{P}) \leq 8$

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Vermoedens:

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Proper colourings

Bewezen grenzen:

$6 \leq \chi_{\mathrm{pCFo}}(\mathcal{P}) \leq 8$

$\chi_{\mathrm{pCFc}}(\mathcal{P}) = 4$

$6 \leq \chi_{\mathrm{pUMo}}(\mathcal{P}) \leq 10$

$6 \leq \chi_{\mathrm{pUMc}}(\mathcal{P}) \leq 8$

Vermoedens

Odd colouring

Bewezen grenzen:		*Vermoedens:*
$5 \leq \chi_{\mathrm{O}}(\mathcal{P}) \leq 8$	$\rightarrow$	$\chi_{\mathrm{O}}(\mathcal{P}) = 5$

Proper colourings

Bewezen grenzen:		*Vermoedens:*
$6 \leq \chi_{\mathrm{pCFo}}(\mathcal{P}) \leq 8$		$\chi_{\mathrm{pCFo}}(\mathcal{P}) = 6$
$\chi_{\mathrm{pCFc}}(\mathcal{P}) = 4$	$\rightarrow$	
$6 \leq \chi_{\mathrm{pUMo}}(\mathcal{P}) \leq 10$		$\chi_{\mathrm{pUMo}}(\mathcal{P}) \leq 7$
$6 \leq \chi_{\mathrm{pUMc}}(\mathcal{P}) \leq 8$		$\chi_{\mathrm{pUMc}}(\mathcal{P}) = 6$

Vermoedens 2

Improper colourings

Bewezen grenzen:

$$4 \leq \chi_{\mathrm{iCFo}}(\mathcal{P}) \leq 5$$

$$3 \leq \chi_{\mathrm{iCFc}}(\mathcal{P}) \leq 4$$

$$5 \leq \chi_{\mathrm{iUMo}}(\mathcal{P}) \leq 10$$

$$4 \leq \chi_{\mathrm{iUMc}}(\mathcal{P}) \leq 6$$

Vermoedens 2

Improper colourings

Bewezen grenzen:		*Vermoedens:*
$4 \leq \chi_{\mathrm{iCFo}}(\mathcal{P}) \leq 5$		$\chi_{\mathrm{iCFo}}(\mathcal{P}) = 4$
$3 \leq \chi_{\mathrm{iCFc}}(\mathcal{P}) \leq 4$	$\rightarrow$	$\chi_{\mathrm{iCFc}}(\mathcal{P}) = 3$
$5 \leq \chi_{\mathrm{iUMo}}(\mathcal{P}) \leq 10$		$\chi_{\mathrm{iUMo}}(\mathcal{P}) = 5$
$4 \leq \chi_{\mathrm{iUMc}}(\mathcal{P}) \leq 6$		$\chi_{\mathrm{iUMc}}(\mathcal{P}) = 4$

- Conjecture van $\chi_{\mathrm{iCFc}}(\mathcal{P})$ is al weerlegd[4].
- $\chi_{\mathrm{iCFc}}(\mathcal{P}) = 4$

[4] Lukáš Málik. *Conflict-free colouring of planar graphs [online]*. Bakalářská práce. SUPERVISOR: prof. RNDr. Petr Hliněný, Ph.D.. 2023 [cit. 2025-11-26]. URL: https://theses.cz/id/u5w3fx/.

Methodologie

- Backtracking
- Houd mogelijke kleuren bij

Methodologie

- Backtracking
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Algorithm 1 Kleuren van grafen

```
1: 𝒢 ← grafen gegenereerd door plantri
2: for G ∈ 𝒢 do
3:     for maxColor ≤ 10 do
4:         if colorGraph(G, maxColor, 0) then
           return maxColor
```

Algorithm 2 `colorGraph`$(G, maxColor, index)$

```
if All vertices colored then return True
vertex ← vertices[index]
for Each available color of vertex do
    col(vertex, color)
    if neighbourhood(vertex) is fully colored ∧ ¬isCorrectlyColored(vertex) then
        continue
    updateNeighbours(vertex, color)
    if colorGraph(G, maxColor, getBestIndex()) then
        return True
uncol(vertex, color)
return False
```

Algorithm 3 `updateNeighbours`$(vertex, color)$

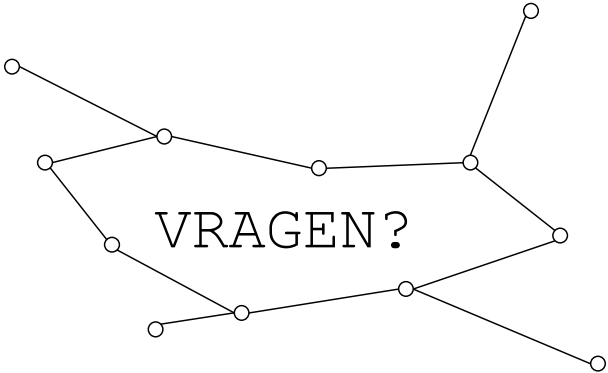
```
for neighbour ∈ neighbourhood(vertex) do
    if neighbourhood(neighbour) is fully colored then
        return isCorrectlyColored(neighbour)
    else if neighbourhood(neighbour) contains only one uncolored vertex then
        removeAvailableColorsForColoring(uncoloredNeighbour)
    else if coloring is proper then
        removeAvailableColor(neighbour, color)
```

Threats to validity

- Incorrect algoritme
- Vergelijken tussen implementaties

Planning

- Optimaliseren
- Patronen in maximaliserende grafen zoeken
- Code omzetten naar C
- Algoritme laten lopen
- Bewijzen correctheid naïef algoritme
- Verslag schrijven


VRAGEN?